Md Siam Hossain Sarker

Software Engineer · Distributed Systems & Identity Infrastructure

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SUMMARY

Software Engineer and M.Sc. Student in Automotive Software Engineering at **TU Chemnitz** with 3+ years building scalable backend systems, distributed services, and identity infrastructure across mining, healthcare, and SaaS domains. Currently at **WebAlive** (Melbourne), architecting **Hykmah** — a multi-product SSO platform operating at the OIDC/PKCE protocol level. **IEEE** member, **Springer & IEEE** published researcher, international hackathon finalist.

EXPERIENCE

Software Engineer

Jan 2026 — Present

WebAlive · Melbourne, Australia (Remote)

- Architecting **Hykmah** — unified SSO platform with custom OIDC/PKCE flows in Keycloak, managing cross-product state, permissions, and auth boundaries.
- Built centralised user provisioning service (FastAPI + MongoDB) for cross-product identity lifecycle and access control.
- Developed backend systems for Exsited and WebCommander in Grails; contributed Umerang backend in C# with GDPR-compliant schema design (merchants, sessions, events).

`Java` `Keycloak` `OIDC/PKCE` `FastAPI` `MongoDB` `Grails` `C#` `MariaDB`

Software Engineer

May 2023 — Dec 2025

Bit Mascot (Pvt.) Ltd. · Dhaka, Bangladesh

- Designed and optimised RESTful APIs across MySQL, PostgreSQL, MongoDB, and DuckDB for high-volume production workloads, following Repository-Service and MVC patterns.
- Built GIS-based spatial query systems for two Australian mining platforms (**Mining Hub**, **Mining Bull**) using PostGIS and spatial indexes for real-time geospatial filtering.
- Implemented async data pipelines (RabbitMQ, Laravel Schedulers) and WebSocket layers to decouple bottlenecks in enterprise workflows.
- Integrated SendGrid, Autobill SDK, and webhook-driven flows across **Locum West**, **Nintendo Repair**, and **Regional Tech Hub**.

`PHP` `Laravel` `CakePHP` `Node.js` `PostgreSQL/GIS` `MySQL` `RabbitMQ` `WebSockets` `DuckDB`

Undergraduate Teaching Assistant

Nov 2021 — May 2023

United International University · Dhaka, Bangladesh

- TA for Structured Programming, OOP, DBMS, and Computer Networks labs; one-on-one mentoring in C/C++, Java, PHP, and MySQL.
- **President — UIU App Forum** (2020–2023): led 800+ member developer community with industry-partnered events on web3, mobile, and career development.

SELECTED PROJECTS

Hykmah Ecosystem

`Keycloak · FastAPI · MongoDB · Grails · C#`

2026 — present

Unified SSO platform with custom OIDC/PKCE flows; single login across independent products with per-product authorisation. Centralised provisioning service manages cross-product access lifecycle.

Log Archiver

`FastAPI · DuckDB · Apache Parquet · RabbitMQ`

2025

Enterprise log archival platform with columnar Parquet storage for efficient analytics queries. RabbitMQ-decoupled ingestion sustains throughput under high load without blocking the write path.

Mining Hub & Mining Bull

`Laravel · Node.js · PostgreSQL/GIS`

↗ mininghub.com

REST APIs and GIS spatial query systems for two Australian mining platforms; real-time commodity tracking with geospatial filtering and spatial indexes.

Reed Graduation

`CakePHP · MySQL · RabbitMQ`

↗ reedevents.com.au

Graduation service platform for universities with CMS for editable certificate templates. Asynchronous certificate generation via RabbitMQ.

Car-Story

`Solidity · Ethereum · React`

↗ github

Decentralised vehicle history on Ethereum smart contracts; removes the need for a trusted central authority. Undergraduate thesis — published at **IEEE TENSYMP 2024**.

EDUCATION

MSc in Automotive Software Engineering

2026 — Present

Technische Universität Chemnitz · Chemnitz, Germany

Faculty of Computer Science · Focus on Advanced Platforms & Distributed Systems

BSc in Computer Science & Engineering

2019 — 2023

United International University · Dhaka, Bangladesh

CGPA **3.90 / 4.00** (Summa Cum Laude, German equiv. **1.15**) · 137 credits · 11 academic scholarships · Thesis published at IEEE TENSYMP 2024

SKILLS

LANGUAGES
Java, Python, TypeScript / JavaScript, PHP, C#, Solidity, C/C++

BACKEND
Laravel, FastAPI, Node.js (Express, NestJS), Grails, CakePHP

IDENTITY & SECURITY
Keycloak, OIDC, PKCE, OAuth2, token enrichment, access control

BLOCKCHAIN
Solidity, Ethereum, smart contract architecture

DATABASES
PostgreSQL / GIS, MySQL, MongoDB, DuckDB, MariaDB

ASYNC & MESSAGING
RabbitMQ, WebSockets, Apache Parquet, event-driven pipelines

FRONTEND
React, React Native, jQuery, Ajax, ES6

TOOLING
Docker, Git, Linux, Postman, Selenium, LaTeX, Jira

PUBLICATIONS

Car-Story: A Decentralised Vehicle Information Storage System Based on Blockchain
IEEE TENSYMP 2024 — Ethereum smart contract architecture for trustless vehicle history verification.

UAVs in Green Health Care for Energy Efficiency and Real-Time Data Transmission
Springer LNNS Vol. 665 · ICISS 2023 — UAV-based healthcare data delivery with MobileNet real-time detection.

HONOURS & ACHIEVEMENTS

International Blockchain Olympiad — Award of Distinction 2023
Top 20 nationally, international finalist · 6th IBCOL, Amsterdam · Team "Unchained".

ICPC Asia Dhaka Preliminary — Honorable Mention (×2) 2020 – 2023
Three appearances across teams Songshoptok, CodeSpeed, and Brute Force.

Hult Prize 2022 — 1st Runner-up (UIU) 2022
Competed at the Islamabad Regional Summit.

CSE Project Show — Runner-up 2021 – 2022
System Analysis & Design Lab (Fall '21), Software Engineering Lab (Spring '22).

CERTIFICATIONS

Python for Everybody Specialization — Coursera / University of Michigan (Jul 2020)
PCAP: Programming Essentials in Python — Cisco Networking Academy (Jul 2020)
Introduction to Cloud Development — edX · HTML, CSS, JavaScript (Apr 2022)
Responsive Web Design — freeCodeCamp (Feb 2022)

LANGUAGES & MEMBERSHIPS

Languages: Bengali (native) · English — Professional proficiency (IELTS 6.0 / B2)
IEEE Student Member · **EMK Center** General Member (since 2018)